Worked solutions to

Bayesian Statistical Methods

With Applications to Machine Learning — Reich & Ghosh (2nd ed.)

Chapter 8

Machine learning

Every exercise is presented in the same order:

- the **exact** exercise text from the book;
- the problem **in plain words**;
- the **background** you need, recalled from scratch;
- the **key idea** that unlocks the problem;
- a **step-by-step solution** — every algebra step written out, with a *Why?* note whenever a step is not obvious;
- a **sanity check** of the answer;
- the boxed **answer**;
- **takeaways** and common mistakes.

All numbers and figures are reproduced by `ch08/code/ch08.R` (base R, splines, BART and causaldata; about 3 minutes, mostly BART).

Sai Sridhar · October 20, 2026

Contents

Answers at a glance 2
Primer: the tools of this chapter 3
Exercise 8.1 — Why the horseshoe is a good shrinkage prior 4
Exercise 8.2 — Horseshoe vs. SSVS for the Gambia data 6
Exercise 8.3 — A spline curve for ozone vs. solar radiation 8
Exercise 8.4 — A heteroskedastic model for ozone 10
Exercise 8.5 — An additive model for ozone 12
Exercise 8.6 — Strengths and weaknesses of five ML approaches 14
Exercise 8.7 — BART tuning parameters for the motorcycle data 15
Exercise 8.8 — Daily bike rentals: BART vs. linear regression 17
Exercise 8.9 — Learning an optimal treatment rule 19
Exercise 8.10 — Do black legislators respond more to black constituents? 21
Exercise 8.11 — Equation learning for an SIR epidemic 23

Answers at a glance

1	$\mathrm{E}(\beta \mid Y) = (1-\kappa)Y$; a half-Cauchy τ gives $\kappa \sim \mathrm{Beta}(\frac{1}{2}, \frac{1}{2})$: "keep or kill".
2	Horseshoe estimates match SSVS (age 0.26, net use -0.25, green 0.29); treated and health centre shrunk toward 0.
3	WAIC picks $J = 4$; clear increasing-then-flat relationship; only 34% of points inside the band for f (expected); not monotone.
4	Variance grows with temperature and falls with wind; heteroskedastic model wins (WAIC 985 vs 1001).
5	Wind and temperature are important and non-linear; solar radiation matters less and is roughly linear.
6	GAM: interpretable but additive; GP: flexible with uncertainty but $O(n^3)$; deep learning: scalable but data-hungry; BART: strong default but black-box; quantile regression: robust, distribution-free tails but no single model.
7	Results are robust to `ntree` and `power`, sensitive to `k`; tune `k` by CV.
8	Year and temperature dominate; BART beats MLR (CV RMSE 651 vs 801), both with $\approx 94\%$ coverage.
9	Optimal rule: $A = 1$ iff $X_2 > 0.5$; T-learner BART matches it for 76% of subjects (S-learner 50%).
10	Black legislators slightly more likely to reply; MATE $\approx +0.05$ $(-0.05,\ 0.16)$: no clear effect.
11	Equation learning recovers $\beta \approx 0.43$, $\gamma \approx 0.27$, $R_0 \approx 1.56$ (truth 0.40, 0.25, 1.60) but its intervals are over-confident.

A five-minute primer on the tools of this chapter

Background you need

1. Shrinkage priors. The *horseshoe* prior $\beta_j \mid \lambda_j, \tau \sim \mathrm{N}(0, \lambda_j^2\tau^2)$, $\lambda_j, \tau \sim$ half-Cauchy(1) shrinks small coefficients strongly toward 0 but leaves large ones almost untouched. A convenient Gibbs representation uses auxiliary inverse-gamma variables: $\lambda_j^2 \mid \nu_j \sim \mathrm{IG}(\frac{1}{2}, 1/\nu_j)$, $\nu_j \sim \mathrm{IG}(\frac{1}{2}, 1)$ (and similarly for τ^2).

2. Basis expansions. Splines write an unknown curve as $f(x) = \sum_j B_j(x)\beta_j$: a non-linear fit by *linear* regression. Additive models (GAMs) add one such curve per covariate.

3. Heteroskedastic regression. Model the variance too: $\mathrm{Var}(Y_i) = \exp(\mathbf{X}_i\boldsymbol{\alpha})$, so e^{α_j} is the multiplicative change in the variance per unit of X_j.

4. BART. $Y = \sum_{m=1}^{M} g(\mathbf{X}; T_m, \mathbf{M}_m) + \varepsilon$: a sum of M (`ntree`) small regression trees, each kept small by a prior. Tuning: `ntree` (number of trees), `power`/`base` (prior probability that a node at depth d splits, base $(1+d)^{-\text{power}}$), and `k` (the leaf values are shrunk so that $\pm k$ prior SDs cover the range of Y; bigger k = smoother fit).

5. Causal effects. With treatment A and confounders $\mathbf{X}$: $g_a(\mathbf{X}) = \mathrm{E}(Y \mid A = a, \mathbf{X})$; conditional effect $g_1 - g_0$; *mixed* (sample-average) treatment effect MATE $= \frac{1}{n}\sum_i \{g_1(\mathbf{X}_i) - g_0(\mathbf{X}_i)\}$. A causal interpretation needs no unmeasured confounding, positivity and consistency.

Exercise 8.1 — Why the horseshoe is a good shrinkage prior

Exercise 1 (exact text from the book)

Say $Y|\beta \sim \text{Normal}(\beta, \sigma^2)$ and $\beta \sim \text{Normal}(0, \tau^2\sigma^2)$.

(a) Prove that the posterior mean of β is $\text{E}(\beta|Y) = \{1 - \kappa(\tau)\}Y$ for $\kappa(\tau) = 1/(1+\tau^2)$.

(b) Discuss how small and large τ lead to different degrees of shrinkage.

(c) Prove that if $\tau \sim \text{HC}(1)$ (as in the horseshoe prior), then $\kappa(\tau) \sim \text{Beta}(1/2, 1/2)$.

(d) Argue that $\kappa(\tau) \sim \text{Beta}(1/2, 1/2)$ is a reasonable shrinkage prior.

In plain words

For one noisy observation of a coefficient, show that the posterior mean shrinks the data by a factor κ, and that the horseshoe's half-Cauchy prior makes κ follow a U-shaped distribution.

Background you need

- Normal–normal: precisions add; posterior mean = precision-weighted average (Chapter 2 primer).
- Half-Cauchy(1) density: $f(\tau) = \frac{2}{\pi(1+\tau^2)}$, $\tau > 0$.
- Change of variables: $f_\kappa(\kappa) = f_\tau(\tau(\kappa)) \left|\frac{\text{d}\tau}{\text{d}\kappa}\right|$.

Key idea

κ is the fraction of the observation that is "shrunk away": $\kappa = 0$ means no shrinkage, $\kappa = 1$ means the estimate is set to 0.

Solution

(a) Posterior mean.

Likelihood precision $\frac{1}{\sigma^2}$; prior precision $\frac{1}{\tau^2\sigma^2}$; prior mean 0. So

$$\text{E}(\beta \mid Y) = \frac{Y/\sigma^2}{1/\sigma^2 + 1/(\tau^2\sigma^2)} = \frac{Y}{1 + 1/\tau^2} = \frac{\tau^2}{1+\tau^2} Y = \Big(1 - \frac{1}{1+\tau^2}\Big)Y = \{1 - \kappa(\tau)\}Y.$$

Why? Multiply numerator and denominator by σ^2, then by τ^2: $\frac{1}{1+1/\tau^2} = \frac{\tau^2}{\tau^2+1}$.

(b) Small vs. large τ.

$\tau \to 0$: $\kappa \to 1$, $\text{E}(\beta \mid Y) \to 0$ — total shrinkage (the prior insists $\beta \approx 0$). $\tau \to \infty$: $\kappa \to 0$, $\text{E}(\beta \mid Y) \to Y$ — no shrinkage (the data speak alone). Intermediate τ gives partial shrinkage, e.g. $\tau = 1$: $\kappa = \frac{1}{2}$, halfway.

(c) Distribution of κ.

Invert $\kappa = \frac{1}{1+\tau^2}$: $\tau = \sqrt{\frac{1-\kappa}{\kappa}}$, and

$$\frac{\text{d}\tau}{\text{d}\kappa} = \frac{1}{2}\Big(\frac{1-\kappa}{\kappa}\Big)^{-1/2} \cdot \Big(-\frac{1}{\kappa^2}\Big) = -\frac{1}{2\kappa^{3/2}(1-\kappa)^{1/2}}.$$

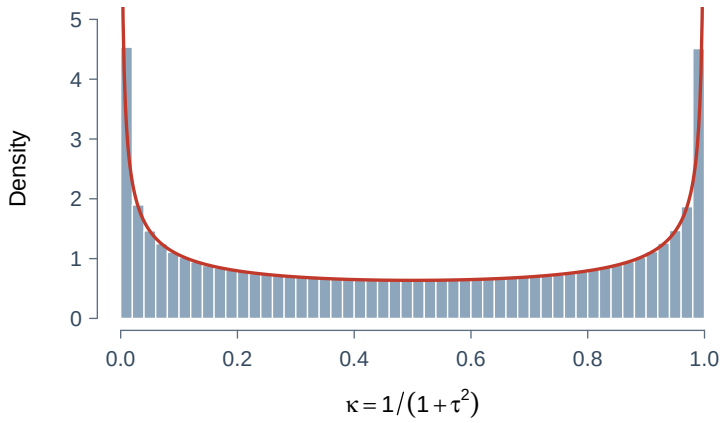
Also $1 + \tau^2 = \frac{1}{\kappa}$, so $f_\tau(\tau) = \frac{2}{\pi}\kappa$. Hence

$$f_\kappa(\kappa) = \frac{2\kappa}{\pi} \cdot \frac{1}{2\kappa^{3/2}(1-\kappa)^{1/2}} = \frac{1}{\pi}\kappa^{-1/2}(1-\kappa)^{-1/2}, \qquad 0 < \kappa < 1,$$

which is the $\text{Beta}(\frac{1}{2}, \frac{1}{2})$ density ($B(\frac{1}{2}, \frac{1}{2}) = \pi$).

Sanity check

Simulation: 10^6 draws of $\tau = |\text{Cauchy}|$ give κ with mean 0.4999 (theory 0.5), $\text{Prob}(\kappa < 0.1) = 0.2051$ (theory 0.2048) and KS distance 0.0008 from $\text{Beta}(\frac{1}{2}, \frac{1}{2})$. ✓



› (d) Why it is a good shrinkage prior.

The $\text{Beta}(\frac{1}{2}, \frac{1}{2})$ density is U-shaped: most mass is near $\kappa = 0$ (*leave the coefficient alone*) and near $\kappa = 1$ (*shrink it to zero*). That mimics a spike-and-slab prior — coefficients are either signal or noise — but continuously, so it is easy to compute with. Large signals escape shrinkage (the heavy Cauchy tail), while noise is shrunk hard.

Answer

(a) $\text{E}(\beta \mid Y) = \frac{\tau^2}{1+\tau^2} Y = (1-\kappa)Y$. (b) Small τ: $\kappa \approx 1$, heavy shrinkage; large τ: $\kappa \approx 0$, none. (c) $f_\kappa(\kappa) = \frac{1}{\pi}\kappa^{-1/2}(1-\kappa)^{-1/2}$, i.e. $\text{Beta}(\frac{1}{2}, \frac{1}{2})$. (d) The U shape says "either keep or kill", the ideal behaviour for sparse signals.

Takeaway & common mistakes

Compare with a $\text{Uniform}(0, 1)$ distribution for κ, which would favour moderate shrinkage of everything — bad for sparse problems.

Exercise 8.2 — Horseshoe vs. SSVS for the Gambia data

Exercise 2 (exact text from the book)

In Section 6.2, we used stochastic search variable selection (SSVS) to analyze childhood malaria data.

(a) Fit the model in (6.5) using horseshoe priors for the regression coefficients, and verify MCMC convergence.

(b) Compare the results using horseshoe regression with the SSVS results in Table 6.2.

(c) In general, what are some pros and cons of SSVS versus horseshoe regression?

In plain words

Redo the book's malaria variable-selection analysis with a horseshoe prior instead of spike-and-slab and compare.

Background you need

Model (6.5): logit $\text{Prob}(Y_i = 1) = \alpha + \sum_{j=1}^{5} X_{ij}\beta_j$ with the five covariates (age, net use, treated net, greenness, health centre) standardised. Table 6.2 (SSVS): inclusion probabilities 1.00, 1.00, 1.00, 0.79, 0.56 and medians $0.26,\ -0.25,\ 0.29,\ -0.13,\ -0.05$.

Key idea

Given the local and global scales, the horseshoe is just a normal prior with variances $\lambda_j^2\tau^2$; the scales themselves have inverse-gamma full conditionals (primer, point 1). Only $\boldsymbol{\beta}$ needs a Metropolis step.

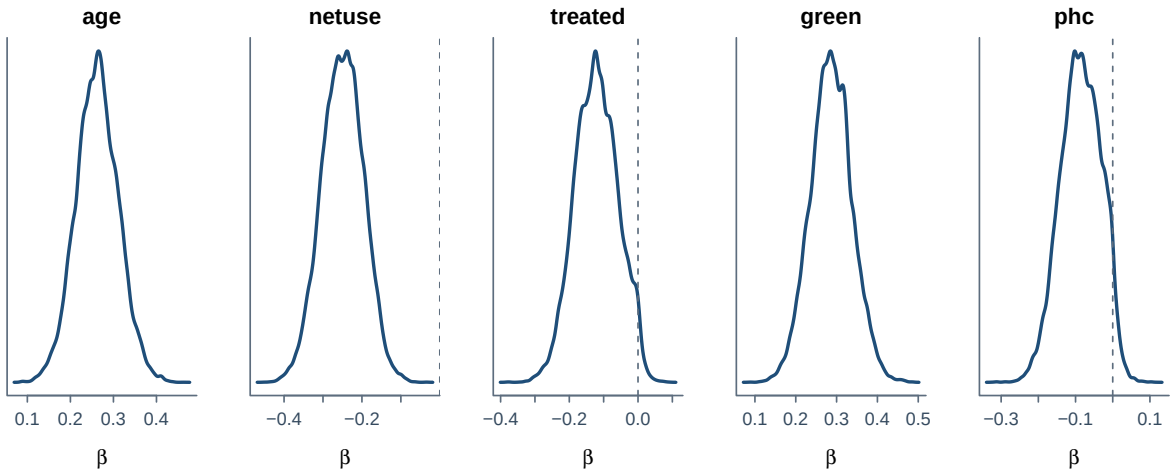
Solution

(a) Sampler and convergence.

Each iteration: block Metropolis for $(\alpha, \boldsymbol{\beta})$ given the scales (acceptance 25%), then Gibbs updates of $\lambda_j^2, \nu_j, \tau^2, \xi$. Three chains $\times$ 20,000 kept draws: all $\hat{R} \leq 1.001$, ESS 2,150–2,670.

(b) Comparison.

	Horseshoe			SSVS (Table 6.2)	
	median	95% interval	GLM (no prior)	incl. prob.	median
age	0.262	(0.166, 0.361)	0.273	1.00	0.26
net use	−0.249	(−0.355, −0.148)	−0.253	1.00	−0.25
treated	−0.122	(−0.241, −0.001)	−0.129	0.79	−0.13
greenness	0.286	(0.188, 0.388)	0.294	1.00	0.29
health centre	−0.085	(−0.193, 0.009)	−0.099	0.56	−0.05



The three strong effects agree almost exactly. For the two weak ones, both methods shrink toward 0: SSVS pushes the health-centre median to -0.05 (because 44% of its posterior mass is exactly 0), the horseshoe to -0.085 (continuous shrinkage; its interval just includes 0). Treated nets: both give about -0.12 with an interval touching 0.

› (c) Pros and cons.

- **SSVS** — pros: gives inclusion probabilities and model probabilities directly; exact zeros make selection explicit. Cons: discrete indicators mix slowly, especially with correlated predictors (Exercise 6.4); results depend on the slab variance; 2^p model space.
- **Horseshoe** — pros: continuous, so standard (or HMC) samplers work and scale to thousands of predictors; few tuning choices; large effects are nearly unbiased. Cons: no exact zeros, so a selection rule must be added (e.g. intervals or κ_j thresholds); heavy-tailed scales can make sampling slow.

Answer

(a) Converged ($\hat{R} \leq 1.001$, ESS $> 2{,}000$). (b) Practically identical conclusions: age, net use and greenness are clearly important; treated nets and health centre are weak and shrunk toward 0 by both priors. (c) SSVS gives explicit selection probabilities but mixes poorly in large/correlated problems; the horseshoe scales well and is easy to sample but needs a separate selection rule.

Takeaway & common mistakes

For five well-separated predictors the choice hardly matters; the differences appear with many, correlated predictors.

Exercise 8.3 — A spline curve for ozone vs. solar radiation

Exercise 3 (exact text from the book)

Download the airquality dataset in R. Fit the model

$$\text{ozone}_i \sim \text{Normal}\{f(\text{solar.R}_i), \sigma^2\}$$

where the function f is approximated using spline basis functions.

(a) Use model selection tools to select the number of spline basis functions.

(b) Make a scatterplot of solar radiation (x-axis) versus ozone (y-axis) and overlay the fitted curve f with a 95% interval for f for each value of solar radiation. Visually, does the estimated curve fit the data well? Can you conclude that there is a statistically significant relationship between these two variables?

(c) Do 95% of the observations fall in the 95% credible interval in part (b)? Is this a concern? If not, why? If so, how would you address this concern?

(d) Looking at the data, does it appear that f is monotonically increasing? How would you modify your analysis to incorporate the restriction that f is a strictly increasing function of solar radiation?

In plain words

Fit a smooth curve of ozone against solar radiation, choose its flexibility, check the uncertainty band, and think about forcing the curve to increase.

Background you need

B-spline regression (Exercise 4.10) and WAIC (Chapter 6). A *credible band for* f describes uncertainty about the *mean*; a *prediction interval* also includes the noise σ.

Key idea

Choose J by WAIC/DIC, then read the band for f and the prediction intervals separately.

Solution

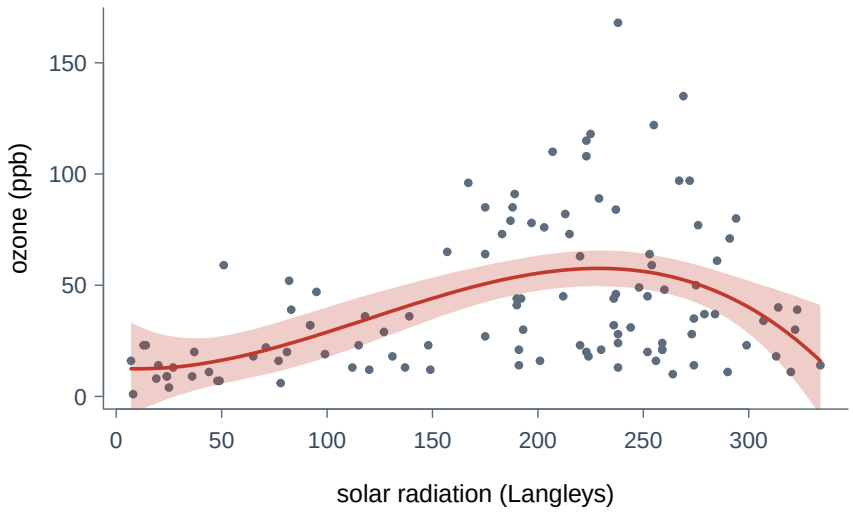
(a) Selecting J

(111 days; cubic B-splines with J basis functions; vague priors):

J	4	5	6	7	8	9	10	11	12
WAIC	**1069.9**	1071.4	1073.5	1073.5	1076.4	1076.5	1077.0	1079.3	1082.0
DIC	**1070.6**	1072.1	1074.4	1074.6	1077.2	1077.2	1078.0	1080.0	1082.6

Both choose $J = 4$ (a single cubic polynomial piece) — the data do not support a wiggly curve.

(b) Fitted curve.



Ozone rises with solar radiation up to about 200–250 Langleys and then levels off or dips. The curve captures the average trend, but the scatter around it is huge. The relationship is clearly real: the posterior of the curve's range $\max f - \min f$ is 51 ppb (95% CI 34–72), and a linear slope is 11.6 ppb per SD (95% CI 5.6–17.8), well away from 0.

› (c) Coverage.

Only **34%** of observations lie inside the 95% band for f — *not* a concern, because that band is for the *mean* ozone at each radiation level, not for individual days. The 95% *prediction* intervals ($f+$ noise) contain **96%** of the observations, as they should. (The real concern is the shape of the noise: it is skewed and grows with the mean, see Exercises 6.2 and 8.4.)

› (d) Monotone?

Not quite: the curve dips at high radiation ($\text{Prob}(f \text{ increasing everywhere} \mid \text{data}) = 0.001$), but the dip is within the noise. To force monotonicity: with B-splines, f is non-decreasing whenever the coefficients are ordered, $\beta_1 \leq \beta_2 \leq \cdots \leq \beta_J$. Reparameterise $\beta_1 = \eta_1$, $\beta_j = \beta_{j-1} + e^{\eta_j}$ (positive increments) and put priors on the η_j; or use a truncated-normal prior restricted to ordered coefficients.

Answer

(a) WAIC and DIC both select $J = 4$. (b) The curve increases then flattens; the relationship is clearly significant (range 34–72 ppb). (c) Only 34% of points are inside the band for f — expected, since it is a band for the mean; 96% are inside the prediction intervals. (d) The estimate is not monotone at the top end; impose monotonicity through ordered B-spline coefficients (positive increments).

Takeaway & common mistakes

Never judge a credible band for a mean function by how many data points it contains — that is the job of prediction intervals.

Exercise 8.4 — A heteroskedastic model for ozone

Exercise 4 (exact text from the book)

Using the airquality data from the previous question with response ozone and solar radiation, wind speed and temperature as the covariates, fit the heteroskedastic linear model

$$Y_i|\boldsymbol{\beta},\boldsymbol{\alpha} \sim \text{Normal}\Big\{\sum_{j=1}^{p} X_{ij}\beta_j,\ \exp\Big(\sum_{j=1}^{p} X_{ij}\alpha_j\Big)\Big\}.$$

(a) Confirm that the MCMC chains have converged.

(b) Summarize the covariate effects on the mean and variance.

(c) Does the heteroskedastic fit better than the usual model with constant variance?

In plain words

Let both the mean ozone and the *spread* of ozone depend on solar radiation, wind and temperature.

Background you need

Given $\boldsymbol{\alpha}$, the model is weighted least squares with weights $w_i = e^{-\mathbf{X}_i\boldsymbol{\alpha}}$, so $\boldsymbol{\beta} \mid \boldsymbol{\alpha}$ is normal: precision $\mathbf{X}^\top\mathbf{W}\mathbf{X} + \mathbf{I}/c^2$. $\boldsymbol{\alpha}$ has no conjugate form (Metropolis).

Key idea

Metropolis-within-Gibbs: exact draw of $\boldsymbol{\beta}$ given $\boldsymbol{\alpha}$; block random walk for $\boldsymbol{\alpha}$ with proposal covariance $\propto 2(\mathbf{X}^\top\mathbf{X})^{-1}$ (the large-sample variance of a log-variance regression).

Solution

(a) Convergence.

Three chains × 20,000 kept draws; acceptance for $\boldsymbol{\alpha}$ 28%; all $\hat{R} \le 1.002$; ESS 1,870–8,580.

(b) Effects

(covariates standardised):

	Mean (β, ppb per SD)		Log variance (α, per SD)	
	mean	95% CI	mean	95% CI
intercept	34.7	(29.5, 40.6)	5.92	(5.66, 6.21)
Solar.R	3.3	(0.4, 6.4)	0.20	$(-0.17,\ 0.55)$
Wind	-2.7	$(-6.4,\ 0.1)$	$\mathbf{-0.64}$	$(-0.92, -0.36)$
Temp	**9.6**	(5.5, 14.4)	**0.54**	(0.08, 0.94)

Why? $e^{0.54} = 1.7$: one SD warmer multiplies the variance by 1.7 (SD by 1.3); $e^{-0.64} = 0.53$: one SD windier roughly halves the variance.

Hot, still days have both higher and *more variable* ozone. Once the variance is modelled, the mean effects are smaller than in the constant-variance fit (Exercise 6.1: temperature 15.8, wind -11.8), because the few very high-ozone hot days no longer dominate the fit.

(c) Comparison.

	lppd	p_W	WAIC
heteroskedastic	-481.8	10.7	**985.0**
constant variance	-493.8	6.9	1001.4

Answer

(a) Converged ($\hat{R} \leq 1.002$). (b) Temperature raises the mean (+9.6 ppb/SD) and the variance (×1.7/SD); wind lowers the variance (×0.53/SD) and probably the mean; solar radiation has a small positive mean effect and no clear variance effect. (c) Yes: WAIC 985 vs. 1001 (difference 16), a substantial improvement.

Takeaway & common mistakes

Heteroskedasticity is common whenever a positive response has a wide range; modelling it changes not just the intervals but also the mean estimates.

Exercise 8.5 — An additive model for ozone

Exercise 5 (exact text from the book)

Using the airquality data, fit the GAM model

$$\text{ozone}_i = f_1(\text{solor.R}_i) + f_2(\text{wind.speed}_i) + f_3(\text{temp}_i) + \varepsilon_i$$

where $\varepsilon_i \overset{\text{iid}}{\sim} \text{Normal}(0, \sigma^2)$ and f_1, f_2 and f_3 are each modeled using B-spline basis expansions with five knots. Which are important predictors of ozone, and which appear to have non-linear effects?

In plain words

Let each covariate have its own smooth curve, then decide which curves matter and which are genuinely curved.

Background you need

GAM: $f_k(x) = \sum_j B_{kj}(x)\beta_{kj}$. Each basis is centred so that the intercept is identifiable. Compare models by WAIC: drop a variable to test *importance*; replace its curve with a straight line to test *non-linearity*.

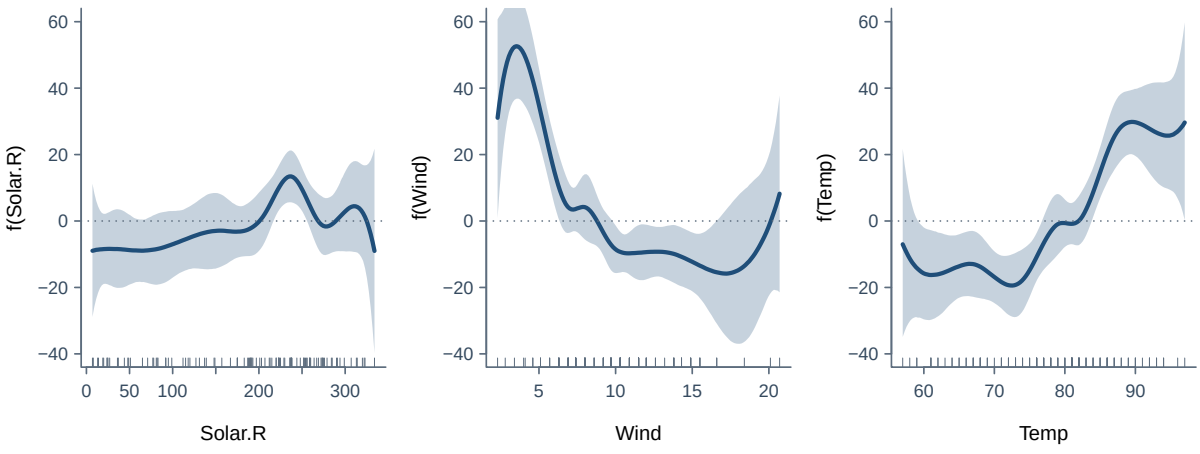
Key idea

Cubic B-splines with 5 interior knots (at the sextiles of each covariate): 8 basis functions per covariate, fitted jointly by the Gibbs sampler for linear regression.

Solution

› Fitted curves

(posterior median and 95% band):



Ozone falls steeply as wind increases from 2 to about 10 mph, then levels off; it rises sharply above about 80°F; solar radiation has a gentler, roughly linear positive effect.

› Model comparison by WAIC.

Model	WAIC	difference from full GAM
full GAM (all smooth)	978.0	0
Solar.R linear, others smooth	**974.2**	-3.8
Wind linear, others smooth	998.1	$+20.1$
Temp linear, others smooth	984.0	$+6.1$
without Solar.R	981.1	$+3.2$
without Wind	1017.4	$+39.4$
without Temp	1015.0	$+37.1$
fully linear model	1001.3	$+23.3$

Why? A large *increase* in WAIC when a curve is made linear means the curvature mattered; a large increase when a variable is dropped means the variable mattered.

Answer

Important: wind and temperature (dropping either costs ≈ 38 WAIC units); solar radiation matters much less (dropping it costs 3). **Non-linear**: wind (making it linear costs 20) and temperature (costs 6); solar radiation's effect is adequately linear (a linear term is even slightly better, -3.8). The additive model beats the linear model by 23 WAIC units.

Takeaway & common mistakes

A good default is "smooth everything, then test which smooths can be simplified".

Exercise 8.6 — Strengths and weaknesses of five ML approaches

Exercise 6 (exact text from the book)

Given a strength and a weakness for each of the following approaches:

(a) Generalized additive modeling

(b) Gaussian process regression

(c) Deep learning

(d) Bayesian additive regression trees

(e) Quantile regression

In plain words

For five modern regression tools, name one thing each does well and one thing it does badly.

Key idea

Consider interpretability, flexibility (non-linearity and interactions), uncertainty quantification and computational cost.

Solution

Approach	Strength	Weakness
(a) GAM	Interpretable: one smooth curve per covariate that can be plotted; fast (linear in the basis coefficients).	Additive: misses interactions unless they are added by hand; knot choice matters.
(b) Gaussian process regression	Very flexible and gives honest, smooth uncertainty bands, especially for interpolation.	Cost grows like n^3 (matrix inversion); kernel choice and high dimensions are hard.
(c) Deep learning	Learns complex features and interactions from huge, high-dimensional data (images, text); scales with GPUs.	Data-hungry and a black box; full Bayesian uncertainty is expensive or approximate.
(d) BART	Excellent default: captures non-linearity and interactions automatically, with uncertainty, and needs little tuning.	Less interpretable; piecewise-constant fits; extrapolates poorly; slower than single-tree methods.
(e) Quantile regression	Describes the whole conditional distribution (e.g. extremes), robust to outliers, no normality assumption.	Each quantile is a separate model (quantiles can cross); a likelihood (e.g. asymmetric Laplace) is needed for a Bayesian version and its uncertainty can be miscalibrated.

Answer

See the table.

Takeaway & common mistakes

No method is best everywhere: choose by data size, need for interpretation, and whether uncertainty quantification matters.

Exercise 8.7 — BART tuning parameters for the motorcycle data

Exercise 7 (exact text from the book)

Fit a semiparametric model to the mcycle data from the MASS package with the time since a motorcycle makes impact as the only predictor and acceleration as the outcome using the wbart function in the BART package.

(a) Describe the role of the tuning parameters ntree, power and k.

(b) Fit the model and plot the fitted values for several values of these tuning parameters. Are the results sensitive to these tuning parameters? If so, which ones?

(c) In practice, how would you select these tuning parameters in a computationally efficient way?

In plain words

Fit BART to the head-acceleration-after-crash curve, vary its three main tuning knobs, and see which matter.

Background you need

BART tuning parameters: primer, point 4.

Key idea

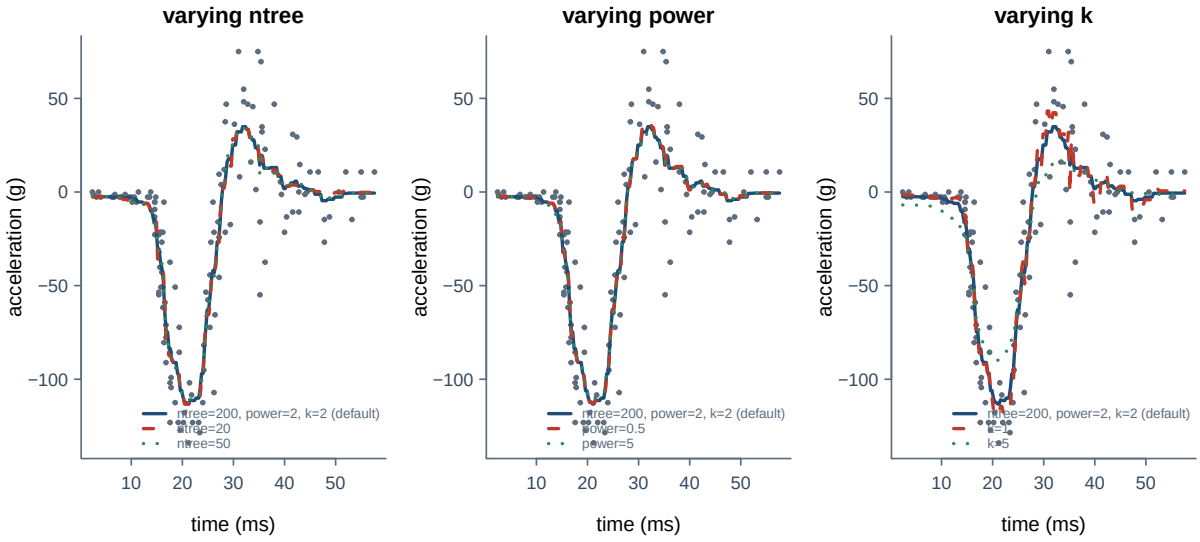
Compare settings by their 5-fold cross-validation RMSE, changing one parameter at a time from the defaults (`ntree`=200, `power`=2, `k`=2).

Solution

(a) Roles.

- `ntree`: number of trees in the sum. More trees = each tree does less (smaller leaf effects), a smoother and more stable fit; too few trees can underfit or mix poorly.
- `power` (with `base`): controls how strongly the prior penalises deep trees ($\text{Prob}(\text{split at depth } d) = \text{base}(1+d)^{-\text{power}}$). Larger `power` = shallower trees = fewer interactions.
- `k`: leaf values are given prior SD $\approx \frac{\text{range}(Y)}{2k\sqrt{\texttt{ntree}}}$. Larger `k` = more shrinkage of the fitted function toward its mean = smoother, less flexible fit.

(b) Fits and sensitivity.



Setting	5-fold CV RMSE	posterior mean σ
default (`ntree`=200, `power`=2, `k`=2)	24.2	22.1
`ntree`=20	24.6	22.0
`ntree`=50	24.6	21.9
`power`=0.5	24.3	22.0
`power`=5	24.2	22.3
`k`=1	25.6	21.2
`k`=5	**27.2**	25.1

The fit is insensitive to `ntree` and `power` (CV RMSE within 0.4). It is *sensitive to* `k`: $k = 5$ over-smooths (it cannot follow the sharp dip after 15 ms; RMSE 27.2), and $k = 1$ under-smooths (a wigglier fit and slightly worse RMSE).

› (c) Efficient selection.

Keep the defaults for `ntree`/`power` (robust) and tune `k` (and possibly the σ prior) by K-fold CV over a small grid, e.g. $k \in \{1, 2, 3, 5\}$, using short chains (a few hundred draws suffice for prediction)„ run in parallel (e.g. the `mc.wbart` function in the BART package runs chains on several cores).

Answer

(a) `ntree`: number of trees (more = smoother, more stable); `power`: penalty on tree depth (higher = shallower trees); `k`: shrinkage of leaf values (higher = smoother function). (b) Results are robust to `ntree` and `power` but sensitive to `k` ($k = 5$ clearly over-smooths). (c) Tune mainly `k` by K-fold CV with short, parallel runs.

Takeaway & common mistakes

BART's defaults are good; when something needs tuning it is usually the amount of shrinkage (`k`).

Exercise 8.8 — Daily bike rentals: BART vs. linear regression

Exercise 8 (exact text from the book)

Download the daily bike sharing data from the book webpage. The response variable is the daily count (cnt) of bike rentals (the first three values are 985, 801 and 1349), which we model as Gaussian.

(a) Describe and justify how you process the other variables to be used as predictors.

(b) Regress cnt onto the predictors from (a), and determine the most important predictors from your model.

(c) Use five-fold cross-validation to compare BART and multiple linear regression in terms of mean squared prediction error and coverage of 95% prediction intervals.

In plain words

Predict daily bike rentals in Washington D.C. (2011–2012) from calendar and weather variables; compare a linear model with BART.

Background you need

Categorical variables must be turned into dummy variables for a linear model (one baseline level dropped). Variables that are deterministic functions of others, or of the response, must be removed.

Key idea

Watch out for **leakage**: `casual` + `registered` = `cnt` exactly, so they must not be predictors.

Solution

(a) Processing.

- **Removed**: `instant` (row index), `dteday` (date string, already encoded by `yr`, `mnth`, `weekday`), `casual` and `registered` (they add up to the response — leakage), `atemp` ("feels-like" temperature, correlation 0.992 with `temp`), and `workingday` (exactly equal to "weekday and not a holiday", so perfectly collinear with `holiday` and the weekday dummies).
- **Categorical → dummies**: `season` (3 dummies), `mnth` (11), `weekday` (6), `weathersit` (2; the rare category 4 does not occur).
- **Kept as numeric**: `yr` (0/1), `holiday` (0/1), `temp`, `hum`, `windspeed` (already scaled to $[0, 1]$).

That gives 27 predictors. For the linear model they are standardised.

(b) Most important predictors.

Linear regression (vague priors), coefficients per SD (rentals per day):

Predictor	mean	95% CI	Predictor	mean	95% CI
year 2012	1010	(953, 1068)	weather 3 (rain/snow)	−331	(−398, −264)
temperature	821	(676, 969)	September	270	(125, 415)
winter (season 4)	678	(528, 828)	wind speed	−227	(−290, −167)
summer (season 2)	386	(229, 539)	weather 2 (mist)	−220	(−291, −148)
fall (season 3)	364	(180, 547)			

BART's variable-inclusion proportions agree: temperature (0.079), year (0.057), bad weather (0.045), season 4 (0.041), humidity, wind speed. So: **year** (system growth), **temperature**, **season** and **weather** are the main drivers. (Residual SD: 771 for the linear model, 547 for BART.)

(c) 5-fold cross-validation.

	RMSE	MSE	coverage of 95% PI
linear regression	801	6.41×10^5	0.938
BART	**651**	$\mathbf{4.23 \times 10^5}$	0.937

Answer

(a) Drop identifiers, leakage variables (`casual`, `registered`) and redundant ones (`atemp`, `workingday`); dummy-code season, month, weekday and weather. (b) Year, temperature, season and weather are the most important predictors. (c) BART reduces the MSE by about 34% (RMSE 651 vs. 801); both have close to nominal (94%) interval coverage.

Takeaway & common mistakes

BART wins because it captures non-linear effects (e.g. temperature helps only up to a point) and interactions (weather × season) automatically.

Exercise 8.9 — Learning an optimal treatment rule

Exercise 9 (exact text from the book)

Simulate 100 observations from (8.29) with response Y, treatment A and potential confounders $\mathbf{X}$ using the following R code:

```
set.seed(919)
X  ← matrix(runif(500),100,5)
A  ← rbinom(100,1,X[,1])
g0 ← 2 + cos(10*X[,3])
g1 ← 1 + 2*X[,2] + cos(10*X[,3])
Y  ← rnorm(100,(1-A)*g0+A*g1,1)
```

(a) Assuming this model is known, what is the optimal decision rule? That is, for an observation with a given value of $\mathbf{X}$, which value (0 or 1) of A would maximize the fitted mean response? Express the optimal rule as an equation, e.g., $g_1 > g_0$ if $X_2 > X_3$.

(b) Fit a Bayesian machine learning model to these data, and summarize the fitted model.

(c) Using the fitted model, for each of the 100 observations in the training data, which value of the treatment maximizes the mean response? For what proportion of the observations does this match the optimal treatment rule from (a)?

(d) Give the main assumptions needed to have a casual interpretation of these results.

In plain words

Treatment helps some subjects and harms others depending on X_2. Recover who should be treated from 100 noisy observations.

Background you need

Causal effects: primer, point 5. **S-learner**: one model $\hat{g}(A, \mathbf{X})$, effect $\hat{g}(1, \mathbf{X}) - \hat{g}(0, \mathbf{X})$. **T-learner**: separate models $\hat{g}_1(\mathbf{X})$ on treated units and $\hat{g}_0(\mathbf{X})$ on untreated units.

Key idea

The treatment effect is $g_1 - g_0 = 2X_2 - 1$: it depends only on X_2. A model must capture that interaction to find the rule.

Solution

(a) Optimal rule.

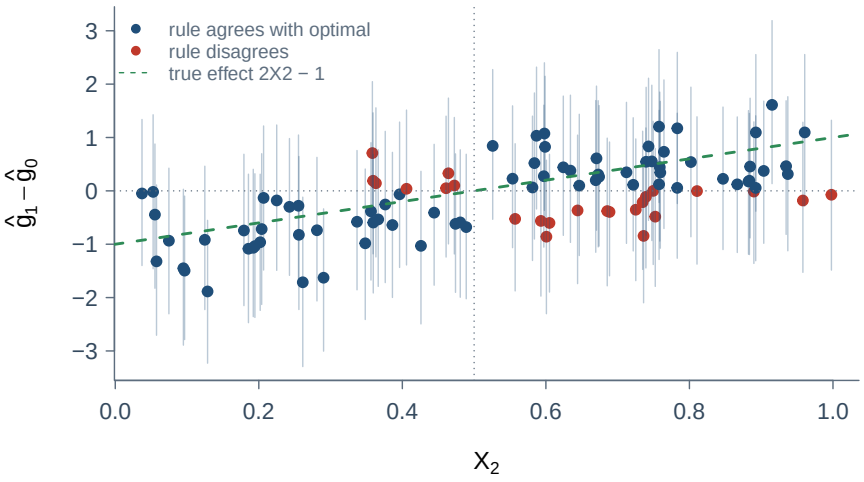
$g_1 - g_0 = (1 + 2X_2 + \cos(10X_3)) - (2 + \cos(10X_3)) = 2X_2 - 1$, so

$$g_1 > g_0 \iff X_2 > 0.5: \qquad \text{treat } (A = 1) \text{ if } X_2 > 0.5, \text{ otherwise } A = 0.$$

(b) Fitted models.

49 of the 100 subjects were treated (treatment depends on X_1 — confounding by design).

- **S-learner** (one BART with A as an input): fits the outcome surface but shrinks the interaction: estimated effects range only from -0.29 to 0.08, although they correlate 0.84 with the truth.
- **T-learner** (two BARTs): effects range from -1.89 to 1.61 (truth -1 to 1), correlation 0.64 with the truth; residual SD ≈ 0.90 in each arm (truth 1). The fitted g_1, g_0 correlate 0.50 and 0.64 with the true functions — with only ≈ 50 noisy points per arm, the $\cos(10X_3)$ wiggle is hard to learn.



› (c) Agreement with the optimal rule.

Treat if the posterior mean of $\hat{g}_1 - \hat{g}_0$ is positive:

Learner	agreement with "$X_2 > 0.5$"
S-learner	50%
T-learner	**76%**

The S-learner's shrunken effects are nearly all on one side of zero, so its rule is no better than a coin toss even though it ranks subjects well; the T-learner recovers the rule for three quarters of the subjects.

› (d) Causal assumptions.

- **No unmeasured confounding**: given $\mathbf{X}$, treatment assignment is independent of the potential outcomes (true here: A depends only on X_1).
- **Positivity**: every subject has a non-zero chance of each treatment, $0 < \text{Prob}(A = 1 \mid \mathbf{X}) < 1$ (here $\text{Prob}(A = 1 \mid \mathbf{X}) = X_1$, which is near 0 or 1 for some subjects).
- **Consistency / SUTVA**: the observed outcome equals the potential outcome under the treatment received, with no interference between subjects.
- The model must be flexible enough to capture g_0 and g_1.

Answer

(a) Treat if and only if $X_2 > 0.5$. (b) BART fits (S- and T-learner); the T-learner captures the effect heterogeneity better. (c) The T-learner's rule matches the optimal rule for 76% of subjects (S-learner 50%). (d) No unmeasured confounding, positivity, consistency/no interference, and a correctly flexible model.

Takeaway & common mistakes

Ranking subjects by benefit and getting the *sign* right are different tasks; shrinkage priors can hurt the latter. With 100 noisy observations, expect imperfect rules.

Exercise 8.10 — Do black legislators respond more to black constituents?

Exercise 10 (exact text from the book)

Load the black_politicians dataset from the causaldata package. To simplify the problem, we will analyze only data with treat_out equal to one. The data consist of the results of emails sent to legislators from black voters from outside the legislator's district. The response variable is the binary indicator of whether the legislator responded to the email, responded. The treatment variable is whether the legislator is black, leg_black, and the remaining 11 variables are used as potential confounders.

(a) Perform a Bayesian logistic regression that includes only the treatment variable as a predictor. Summarize the treatment effect. What are some potential sources of bias in this analysis?

(b) Perform a Bayesian logistic regression that includes both the treatment variable and the confounders (as linear terms), and summarize the treatment effect. Are the results affected by including the potential confounders?

(c) Perform a semiparametric Bayesian logistic regression using the function lbart in the package BART. Compute the posterior distribution of the mixed average treatment effect (MATE), and summarize the results.

In plain words

Emails from (out-of-district) black voters were sent to state legislators. Did black legislators reply more often than other legislators, and does that survive adjustment for district characteristics?

Background you need

Logistic regression and block Metropolis (Chapter 4 primer). MATE: primer, point 5, computed draw-by-draw as $\frac{1}{n}\sum_i\{\hat{p}_i(A=1)-\hat{p}_i(A=0)\}$.

Key idea

Black legislators represent very different districts (far more black residents, more Democratic, more Southern), so the unadjusted comparison mixes the legislator's race with district characteristics.

Solution

Data.

2,779 emails; response rate 31.8% for black legislators (179 emails) vs. 28.7% for others.

(a) Treatment only.

log OR 0.15 $(-0.19,\ 0.47)$, OR 1.16, $\text{Prob}(\text{OR}>1)=0.81$: a small, uncertain positive difference. *Possible biases*: confounding by district demographics (black legislators' districts differ greatly — standardised mean difference 2.98 for % black, 0.96 for Democratic party, 0.62 for South), by legislator characteristics (party, chamber, staff resources, professionalism of the legislature), and by email-handling practices.

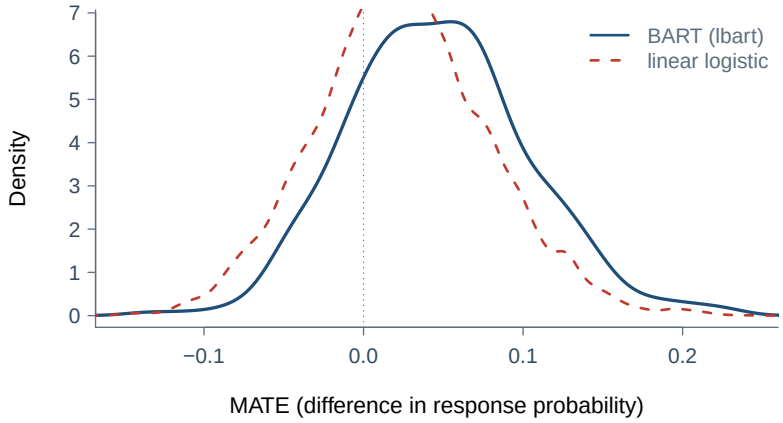
(b) With the 11 confounders

(standardised linear terms): log OR 0.11 $(-0.44,\ 0.62)$, OR 1.12, $\text{Prob}(\text{OR}>1)=0.67$. The point estimate barely moves but the interval widens by $\approx 60\%$: the treatment is strongly associated with the confounders, so adjusting removes information. Strong confounder effects: legislature professionalism (Squire index, $+0.21$), senators ($+0.19$), Democrats (-0.18), larger districts (-0.14).

(c) BART (`lbart`).

Predict each email's response probability with `leg_black` set to 1 and to 0, average the difference over the sample, draw by draw:

	MATE (difference in response probability)	95% CI
naive difference	0.032	—
linear logistic (b)	0.024	$(-0.081,\ 0.134)$
BART	0.046	$(-0.051,\ 0.161)$



$\text{Prob}(\text{MATE} > 0) = 0.79$ under BART.

Answer

(a) OR 1.16 $(0.83,\ 1.60)$: a weak, uncertain advantage for black legislators, potentially biased by large differences in their districts. (b) Adjusting for the confounders leaves the estimate similar (OR 1.12) but much less certain. (c) BART MATE $= +0.05$ (95% CI -0.05 to 0.16), $\text{Prob}(\text{MATE} > 0) = 0.79$: black legislators may be somewhat more responsive to out-of-district black constituents, but the data do not establish it.

Takeaway & common mistakes

When treated and control groups differ this much on confounders (poor overlap), adjustment widens intervals — an honest reflection of how little the data can say.

Exercise 8.11 — Equation learning for an SIR epidemic

Exercise 11 (exact text from the book)

Use the R package covid19.analytics to generate data from the SIR model similar to (8.6.5) for Uruguay as

```
data ← covid19.data("ts-confirmed")
SIR  ← generate.SIR.model(data,"Uruguay",
          tot.population=3500000)
```

Assume that you observe noisy (with multiplicative gamma errors) versions of the number of susceptible and infected,

```
set.seed(1)
S ← SIR$model$S*rgamma(90,100,100)
I ← SIR$model$I*rgamma(90,100,100)
```

Fit a Bayesian equation learning model that regresses $dI(t)/dt$ onto $S(t)$ and $I(t)$, and summarize the disease dynamics you learn from the data.

Note

`covid19.analytics` downloads live (and now discontinued) Johns Hopkins data feeds, so its output is not reproducible. We therefore generate the SIR trajectory directly: population $N = 3.5$ million, 50 initial infections, transmission rate $\beta = 0.40$ and recovery rate $\gamma = 0.25$ per day ($R_0 = 1.6$), 90 days, solved by 4th-order Runge–Kutta. The noise is added exactly as in the exercise (`set.seed(1)`; $\times \text{Gamma}(100, 100)$, i.e. about 10% multiplicative noise).

In plain words

From noisy daily counts of susceptible and infected people, recover the equation that governs the epidemic, and its transmission and recovery rates.

Background you need

SIR model: $\frac{\mathrm{d}S}{\mathrm{d}t} = -\beta\frac{SI}{N}$, $\frac{\mathrm{d}I}{\mathrm{d}t} = \beta\frac{SI}{N} - \gamma I$. Dividing by I:

$$\frac{\mathrm{d}\log I}{\mathrm{d}t} = \beta\,\frac{S}{N} - \gamma,$$

which is *linear* in (β, γ). $R_0 = \beta/\gamma$ is the average number of people one infected person infects in a fully susceptible population.

Key idea

Equation learning in two steps: (1) estimate the derivative from the data by smoothing; (2) regress it on a *library* of candidate terms and see which terms are needed. Working with $\log I$ turns the multiplicative noise into additive noise.

Solution

› Step 1 — derivatives.

Smooth $\log I(t)$ and $\log S(t)$ with smoothing splines (smoothness chosen by generalised cross-validation: 8.5 and 6.9 effective degrees of freedom) and differentiate the smooth of $\log I$.

› Step 2 — library regression.

Regress $\frac{\mathrm{d}\log I}{\mathrm{d}t}$ on $\{1,\ S/N,\ I/N\}$ (vague priors; response in % per day so the variance prior is genuinely vague):

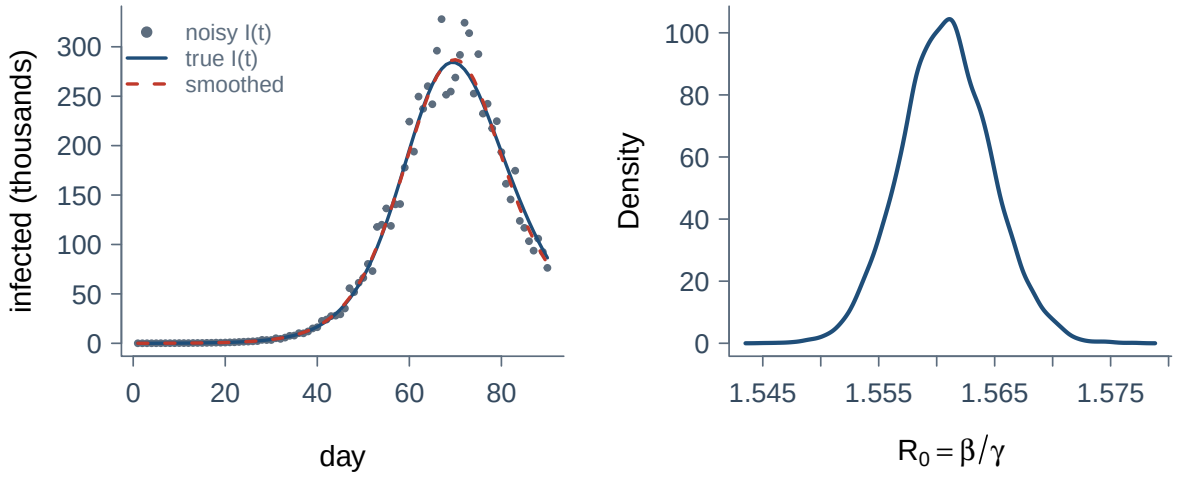
Term	coefficient	95% CI
constant ($=-\gamma$)	-0.272	$(-0.279,\ -0.265)$
S/N ($=\beta$)	0.426	(0.419, 0.433)
I/N (not in SIR)	-0.016	$(-0.068,\ 0.034)$

The extra term I/N is clearly unnecessary (its interval comfortably includes 0): the data "learn" the SIR structure.

› Step 3 — reduced model

$\frac{\mathrm{d}\log I}{\mathrm{d}t} = \beta\frac{S}{N} - \gamma$:

	posterior mean	95% CI	truth
β	0.428	(0.423, 0.433)	0.40
γ	0.274	(0.270, 0.278)	0.25
$R_0 = \beta/\gamma$	1.561	(1.553, 1.569)	1.60



› Dynamics learned.

Each infected person infects ≈ 0.43 others per day and recovers after $1/\gamma \approx 3.7$ days on average; $R_0 \approx 1.56 > 1$, so the epidemic grows until enough people are immune ($S/N < 1/R_0 \approx 0.64$), then declines. Infections peak around day 69 (about 284,000 infected) and 60% of the population has been infected by day 90.

Sanity check

The estimates are close to the truth (all within 10%), but the 95% intervals are *too narrow* and do not contain the true β and γ. Two reasons: (i) the regression treats the smoothed derivatives as independent data, although smoothing makes their errors strongly correlated; (ii) early in the epidemic $S/N \approx 1$, so β and γ are nearly confounded (posterior correlation 0.97) and only their difference (the growth rate) and ratio are well determined. A fully Bayesian alternative is to fit the ODE solution directly to the noisy counts (an ODE likelihood), which propagates all uncertainty.

Answer

The learned equation is $\frac{\mathrm{d}I}{\mathrm{d}t} \approx 0.43\frac{SI}{N} - 0.27I$ (an extra I^2-type term is not supported): transmission $\beta \approx 0.43$/day, recovery $\gamma \approx 0.27$/day (infectious period ≈ 3.7 days), $R_0 \approx 1.56$. Estimates are close to the truth (0.40, 0.25, 1.60), but the two-step method's intervals are over-confident.

Takeaway & common mistakes

Equation learning can recover the *form* of the dynamics from noisy data; for honest *uncertainty* about the rates, fit the differential equation itself.